Exploratory Data Analysis and Visualization

Social Media & Mental Health

Name: Chengxu FAN

ID:72540367

Date: 01/12/2025

Ⅰ. Introduction

In modern society, social media has became a favored platform for people, whether for daily communication, sharing new information, or staying up-to-date with the news. The use of social media has occupied a large part of people's time.

In recent years, numerous studies have examined the impact of social media on individuals' psychological well-being. For example, Blanchard, L. et al. studied the impact of social media on body image, self-esteem and stress in teenagers. They found that the duration of social media use was positively correlated with depressive, disordered eating symptoms and body dissatisfaction [1]. Alonzo, R. et al. pointed out that long-term use of social media by teenagers would affect sleep quality and was a risk factor for poor mental health [2]. A large number of studies have shown that long-term use of social media has adverse effects on mental health. However, most of these studies focus on teenagers, whose mental health is easily affected by external factors. Therefore, does the duration of social media use have an impact on mental health across all age groups?

This project examines the relationship between mental health and the duration of social media use encompassing individuals of all ages to determine the outcome. The impact of social media is evaluated from aspects such as sleep quality, stress level, and happiness index.

Ⅱ. Data Description

The dataset is obtained from Kaggle [3], which includes seven numerical variables (age, daily_screen_time, sleep_quality (1-10), stress_level (1-10), days_without_social_media, exercise_frequency, and happiness_index (1-10)), and two categorical variables (gender and social_media_platform).

In the data cleaning process, the data with the gender marked as "other" were removed. Then, instead of using mean, median, or other substitution methods, the data with missing values were

directly deleted to ensure the authenticity of the data distribution. Eventually, 487 valid data entries were obtained.

This project aims to assess the relationship between mental health and social media usage among individuals of different age groups. The age and gender distribution of the cleaned dataset is shown in the following figure.

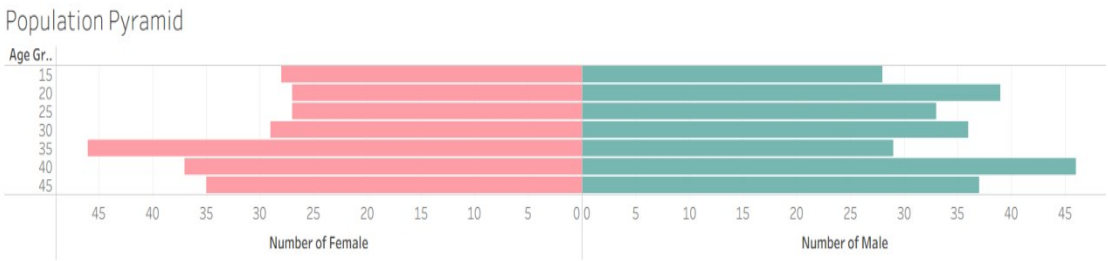


Figure 2.1 Distribution of different ages

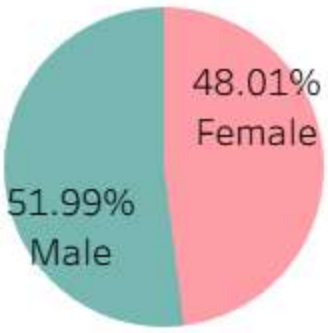


Figure 2.2 Distribution of gender

Ⅲ. Analysis and Visualization

This project conducts a visual analysis of the relationship between the duration of social media usage and, sleep quality, stress level and happiness index, which are main focuses of this project.

The analysis of sleep quality is shown in Figure 3.1. It can be seen from the figure that the duration of social media usage has a significant impact on sleep quality, with a severe decline from around 10 points for users with low usage duration to around 3 points for those with long usage duration.

This indicates that the use of social media does indeed affect sleep quality. Possible reasons include the effect of screen light from electronic devices on melatonin and the excitation of the nervous system, which in turn affects sleep. According to the research by Chang, A. M., Aeschbach, D., et al., using electronic devices before bedtime, the blue light emitted by electronic devices can inhibit melatonin, thereby prolonging the time needed to fall asleep and shortening the rapid eye movement period during the sleep cycle, which reduces sleep quality both before and during sleep [4]. According to the research by Yongzhi Jiang, Xiaoli Bai, et al., excessive use of social media can lead to cognitive arousal before sleep, including various thoughts and emotional arousal, which in turn affects sleep quality [5].

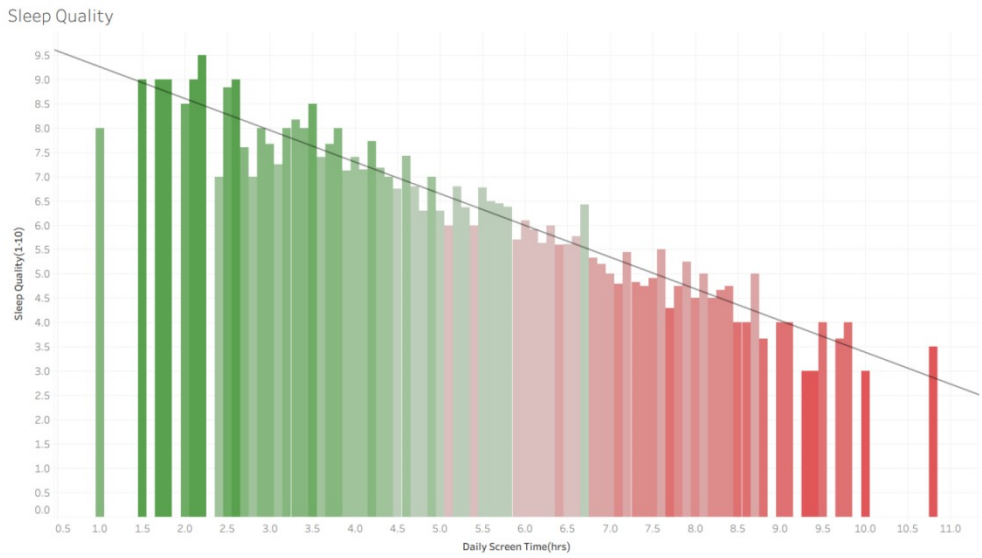


Figure 3.1 Impact of the duration of social media usage on sleep quality

The visualization analysis of the stress index is shown in Figure 3.2. According to the figure, prolonged use of social media will significantly increase an individual's stress level. This may be because social media not only contains positive messages but also negative ones. These messages will affect the stress of users. According to the research of Vogel E. A., Rose J. P., et al., using social media will prompt people to develop a competitive mentality, and continuous upward comparisons will eventually lead to damage to self-esteem and an increase in individual stress [6].

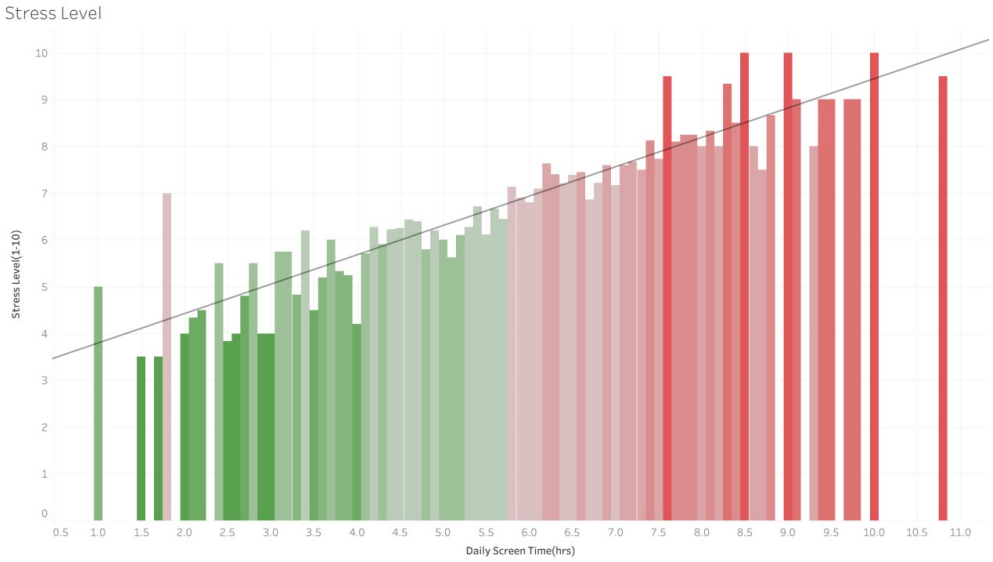


Figure 3.2 Impact of duration of social media usage on stress level

The analysis of the happiness index is shown in Figure 3.3. Similarly, prolonged use of social media also affects an individual's assessment of happiness. However the impact indicators are not as severe as those related to stress and sleep quality (the general happiness index is greater than or equal to 5).

The possible reasons for this result include: regret emotions caused by alternative experiences and time waste, cyberbullying, and negative interactions. These negative interactions can indirectly cause a decrease in an individual's happiness index.

However, social media can also maintain relationships among people and provide positive experiences, which may be the reason why users who have been using social media for a long time do not experience as significant a decline in their happiness index as they do in terms of sleep quality and stress levels.

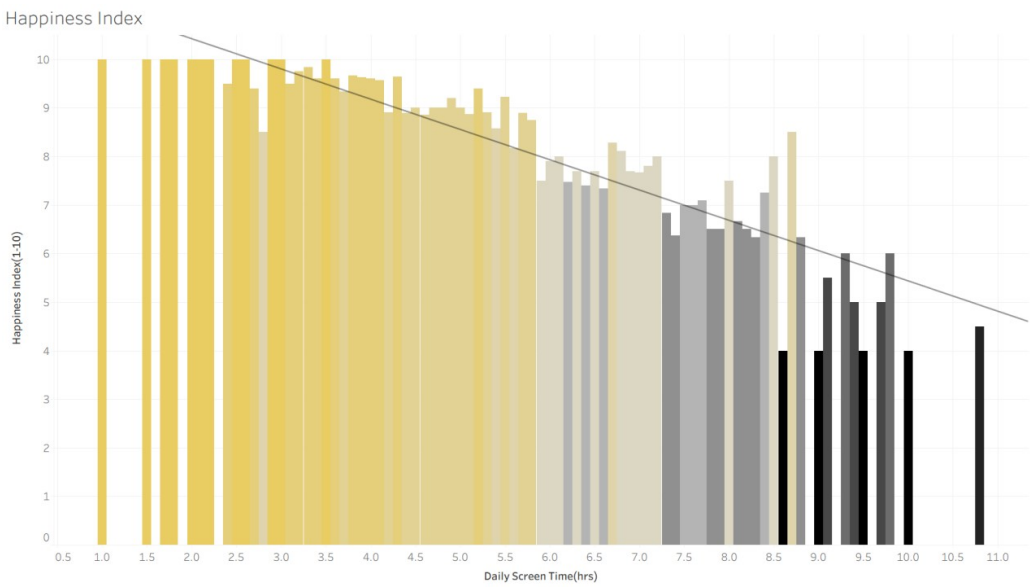


Figure 3.3 Impact of duration of social media usage on happiness index

From the visual results in the above figures, it can be observed that users of all age groups experienced a decline in mental health as the usage time of social media increased. The correlation analysis of the dataset also demonstrated this point. The duration of social media usage shows a strong correlation with sleep quality, stress level, and happiness index. As shown in Figure 3.4.

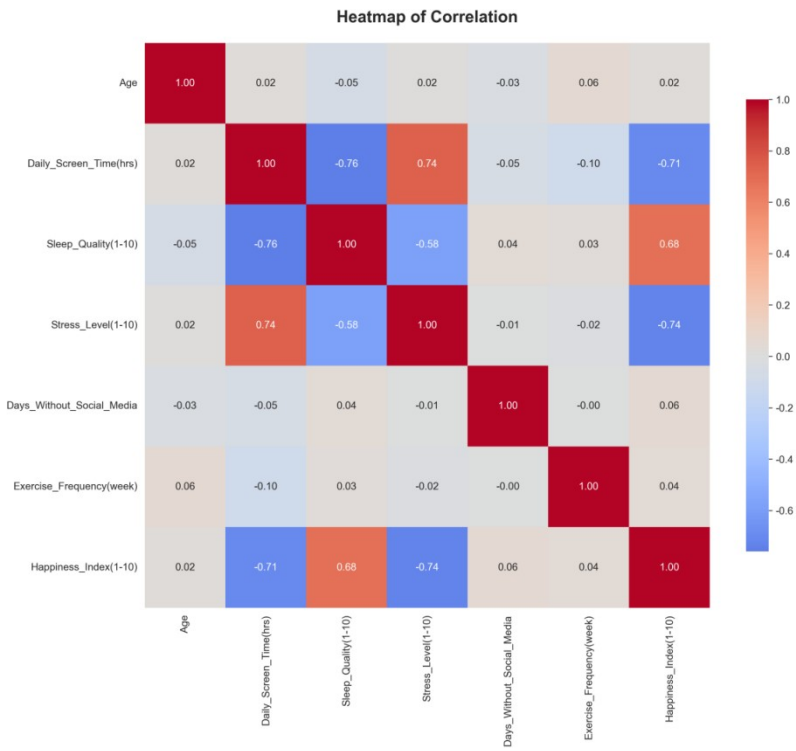


Figure 3.4 Correlation of each variable

Ⅳ. Extra Visualization

Additional interesting phenomena were discovered during the visual analysis of this dataset. Users of different genders tend to choose different social media platforms, as shown in Figure 4.1. Male users prefer to use Instagram and Twitter, while female users prefer to use Facebook. This may be related to the different types of content primarily promoted by each platform. For example, Twitter focuses on political content while Facebook focuses on daily sharing.

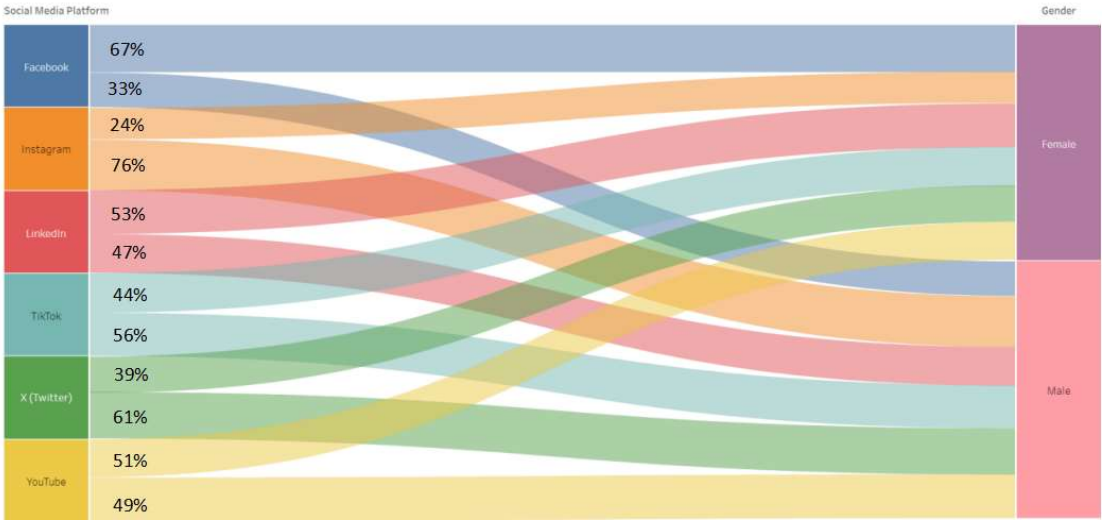


Figure 4.1 Composition of users on different platforms

Additionally, a visual analysis was conducted to examine the stress levels of users across different platforms. As shown in Figure 4.2, it is evident that there is a significant increase in stress levels on both Facebook and Instagram, indicating notable issues in the social environment of these platforms.

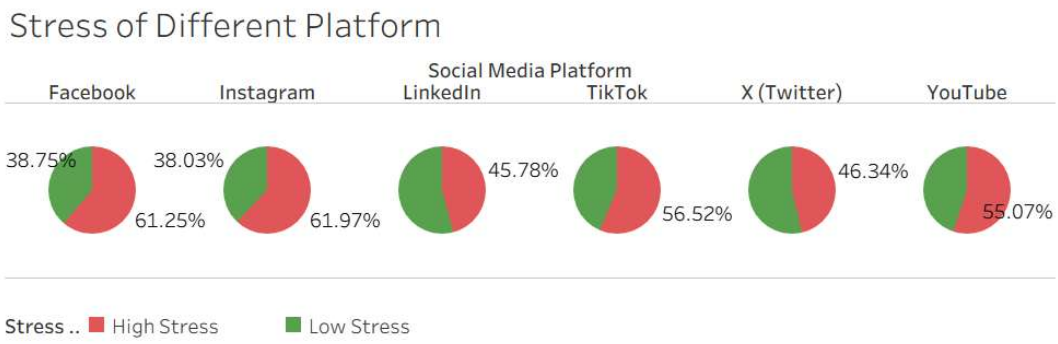


Figure 4.2 Stress of different platforms

Ⅴ. Discussion

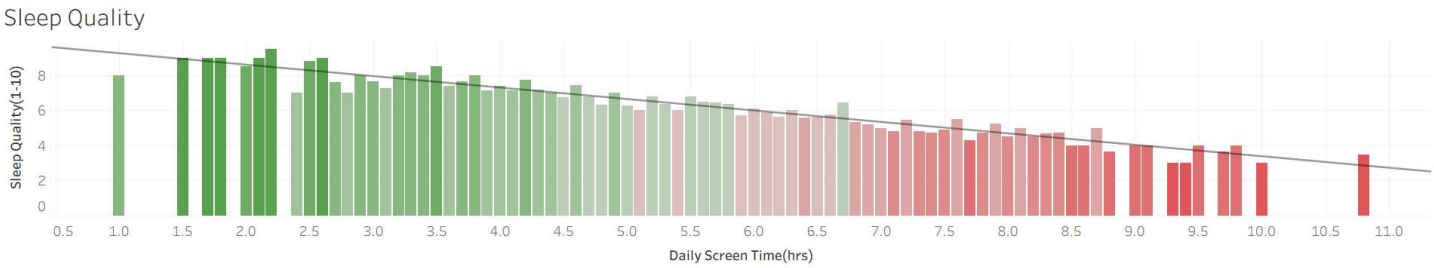
Through the visual analysis in this project, it is evident that the prolonged use of social media has a negative impact on the mental state of people across all age groups. The longer the usage time, the lower the sleep quality and happiness index, and the higher the stress level.

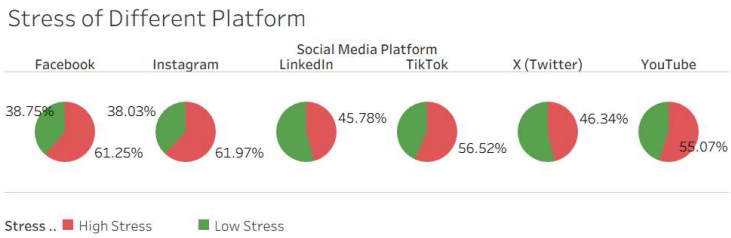
These analysis results can be applied in the fields of psychotherapy and individual mental health counseling. For instance, by analyzing the duration of an individual's social media usage, a comprehensive assessment of the individual's mental state can be made. This is a positive approach to managing mental health issues by reducing social media use and increasing time spent in the real world, focusing on the present and oneself.

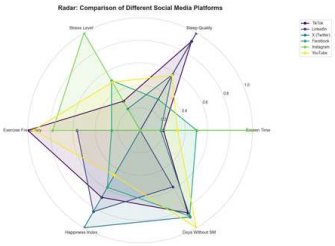
Ⅵ. Conclusion

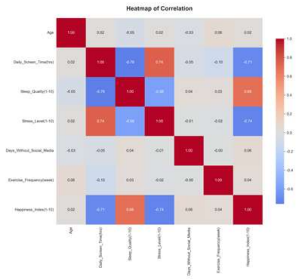
This project conducts a visual analysis of the "Social Media and Mental Health Balance" dataset from Kaggle, and concludes that prolonged use of social media has a negative impact on an individual's mental health. It advocates that people should reduce their reliance on social media and replace its use with longer investments in the real world, in order to improve sleep quality, happiness index, and reduce stress levels.

Ⅶ. Dashboard









Ⅷ. Reference

[1] Blanchard L., Conway-Moore K., Aguiar A., et al. Associations between social media, adolescent mental health, and diet: A systematic review. OBESITY REVIEWS. 2023.

[2] Alonzo R., Hussain J., Stranges S., et al. Interplay between social media use, sleep quality, and mental health in youth: A systematic review. SLEEP MEDICINE REVIEWS. 2021.

[3] Ayesha I.. Social media and mental health balance. Kaggle. 2025. https://www.kaggle.com /datasets/ayeshaimran123/social-media-and-mental-health-balance.

[4] Chang A. M., Aeschbach D., et al. Evening use of light-emitting eReaders negatively affects sleep, circadian timing, and next-morning alertness. PNAS. 2015.

[5] Yongzhi J., Xiaoli B., et al. The relationship between college students' mobile phone dependence and sleep quality: a mediation and moderation effect analysis. Mental Development and Education. 2016.

[6] Vogel E. A., Rose J. P., Roberts L. R., & Eckles K. Social comparison, self-esteem, and social media: A meta-analysis. Psychology of Popular Media Culture. 2014.